

HERO scorecard (vertical) — value ± BCa 95% CI · colour = per-metric min-max (green = best) · * = Δ-vs-frozen CI excludes 0
9 metrics × 8 encoders · WINNER col + BLUE BOX = surgery-vs-pretrain champion duel (tie if within ε) · 10 K bootstrap

	g-frozen	g-pre-2X	g-pre-enc	g-pre-hd	g-s3DI-enc	g-s3DI-hd	g-sNoDI-enc	g-sNoDI-hd	WINNER
action_top1 ↑ better	42.411 ±2.247	48.219 ±2.274*	46.411 ±2.274*	42.740 ±2.329	47.781 ±2.274*	46.466 ±2.283*	46.630 ±2.301*	47.123 ±2.247*	ViT-g pretrain_2X encoder
motion_cos ↑ better	0.007 ±0.001	0.044 ±0.002*	0.042 ±0.002*	0.011 ±0.001*	0.044 ±0.002*	0.050 ±0.002*	0.057 ±0.002*	0.063 ±0.002*	ViT-g surgery noDI head
taxonomy_f1 ↑ better	0.806 ±0.016	0.778 ±0.017*	0.785 ±0.017*	0.795 ±0.017	0.780 ±0.017*	0.786 ±0.017*	0.781 ±0.017*	0.783 ±0.017*	ViT-g pretrain head
future_mse ↓ better	0.637 ±0.001	0.639 ±0.001*	0.632 ±0.001*	0.637 ±0.001	0.628 ±0.001*	0.632 ±0.001*	0.636 ±0.001	0.632 ±0.001*	ViT-g surgery 3stage_DI encoder
rollout ↓ better	0.007 ±0.000	0.005 ±0.000*	0.005 ±0.000*	0.007 ±0.000	0.006 ±0.000*	0.005 ±0.000*	0.005 ±0.000*	0.005 ±0.000*	tie
causal ↓ better	0.655 ±0.002	0.657 ±0.002*	0.652 ±0.002*	0.655 ±0.002	0.650 ±0.002*	0.652 ±0.002*	0.658 ±0.002*	0.652 ±0.002*	ViT-g surgery 3stage_DI encoder
tdist ↓ better	0.007 ±0.000	0.009 ±0.000*	0.008 ±0.000*	0.007 ±0.000	0.009 ±0.000*	0.008 ±0.000*	0.009 ±0.000*	0.008 ±0.000*	tie
teacher_free ↓ better	0.044 ±0.001	0.039 ±0.001*	0.038 ±0.001*	0.044 ±0.001	0.040 ±0.001*	0.038 ±0.001*	0.037 ±0.001*	0.038 ±0.001*	ViT-g surgery noDI encoder
maskratio ↓ better	0.065 ±0.001	0.064 ±0.001*	0.065 ±0.001	0.065 ±0.001	0.066 ±0.001*	0.065 ±0.001	0.062 ±0.001*	0.065 ±0.001	ViT-g surgery noDI encoder